

FPGA PCB and LDO update

Howard Wang and Zitong

Main 35T component choices, compact PCB candidate and direct LDO carrier

Current board list

Board	Latest material	Engineering status
Custom 35T FPGA	M1, 41 × 41 mm, 287 parts	Two-sheet circuit and placement. PCB unrouted.
Existing ASIC LDO	Original nine-regulator board	23 direct digital paths. Actual rails and fitted assembly need verification.
Direct XEM8305 carrier	New Board_Howard Desktop package	Received draft: 59 DRC findings, 20 assigned opens.
Earlier references	R39 main PCB and carrier Step 8	Preserved separately for comparison.

Zitong’s FPGA component list

Function	Exact part	M1 choice
FPGA	XC7A35T-2CSG325I	Keep
125 MHz LVDS clock	AX3DCF1-125.0000T3	Adopt on 1.8 V
Clock splitter	LMK1D2102LRGTR	Omit provisionally
128 Mbit boot flash	IS25WP128F-JKLE	Adopt on 1.8 V, SPI x1
Flash translator	NVT4858HKZ	Adopt, boot validation open

Zitong reports 150 user I/O and four GTP lanes. The stated 122-unit stock is a dated report, not a current purchasing check.

Clock and boot decisions

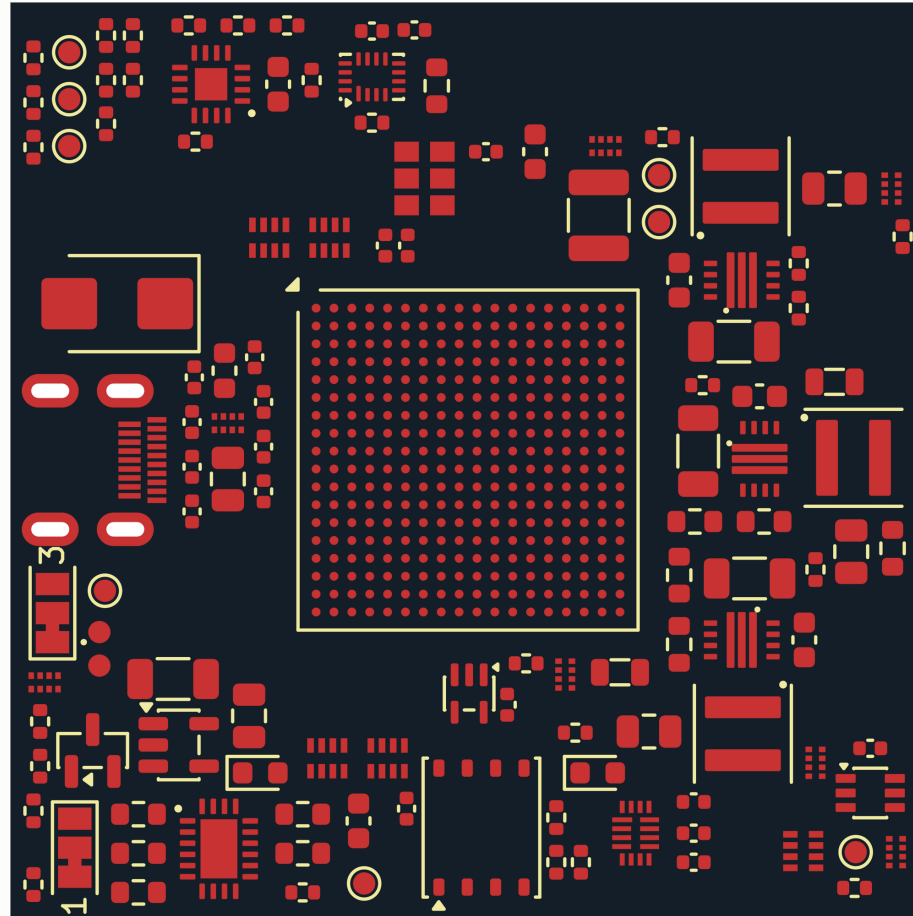
The selected oscillator is 125 MHz LVDS at 1.8 V. M1 keeps the AC-coupling pair and uses a supplier-derived land pattern.

M1 omits the LMK splitter while the GTP ODIV2 fabric-clock path awaits Vivado verification.

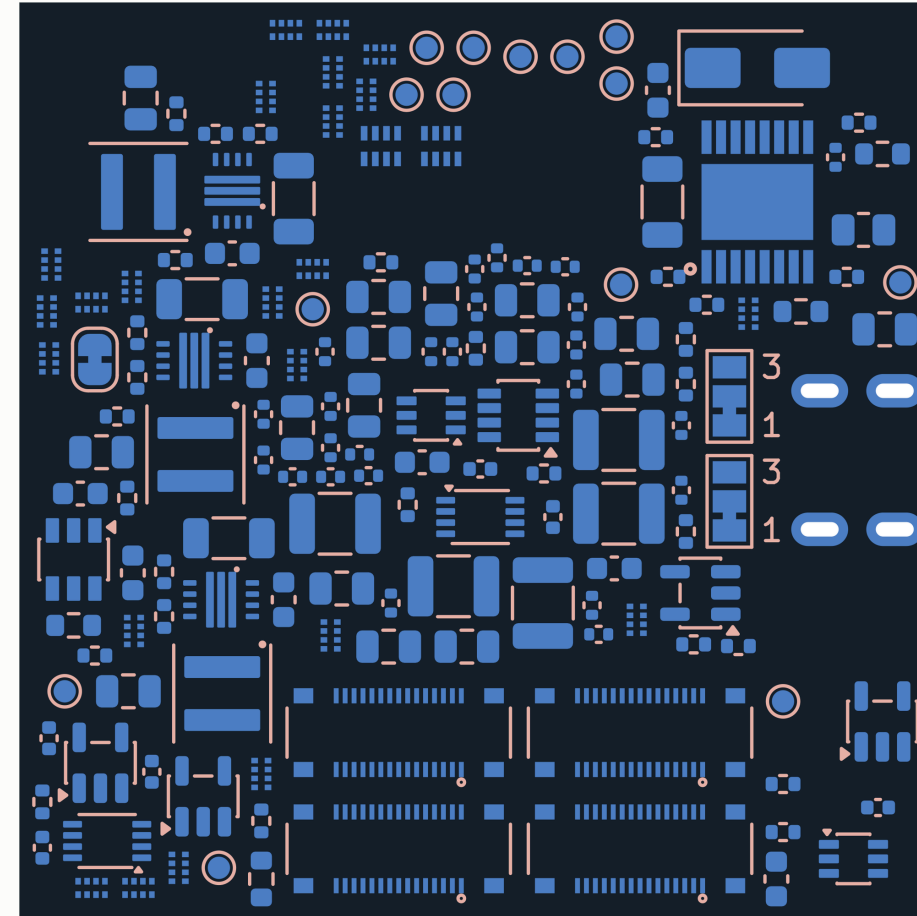
The original slide asks for better than ± 20 ppm but lists a ± 25 ppm oscillator. That requirement still needs reconciliation.

The 1.8 V flash crosses the 1.5 V FPGA configuration interface through NVT4858. SPI x1 boot and brownout behavior need testing.

M1 FPGA PCB in KiCad



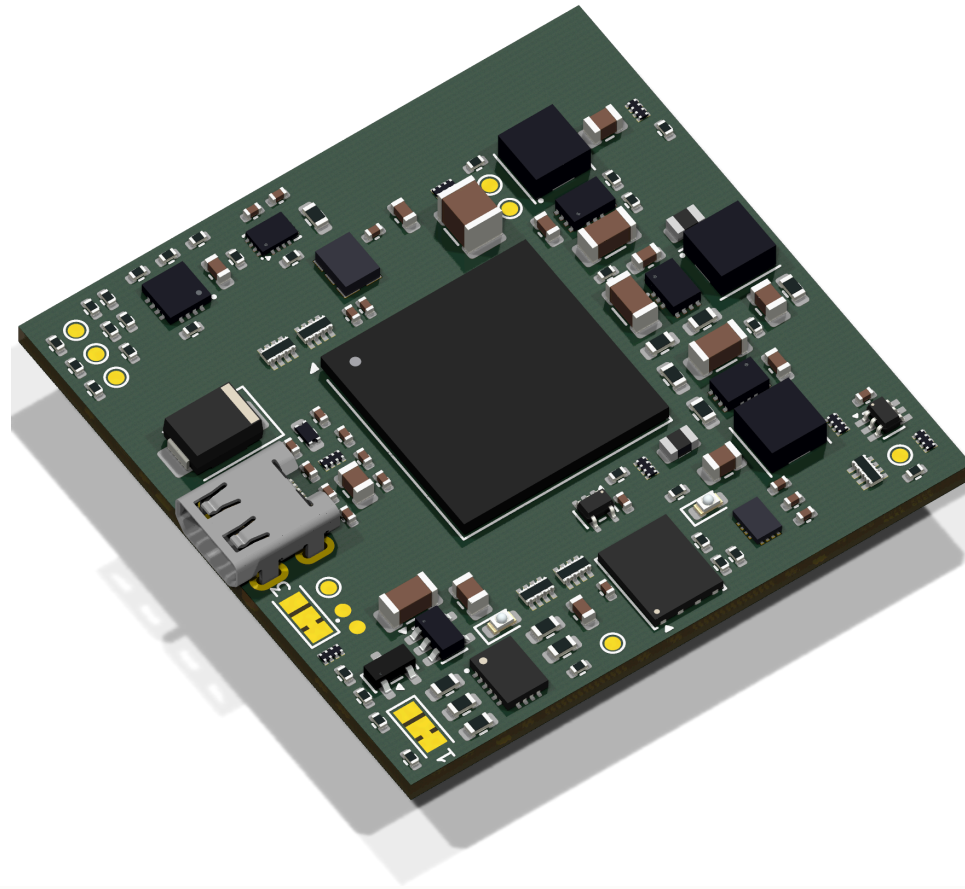
Top copper, pads, silkscreen and outline



Underside with four grouped floating connectors

Measured outline 41 × 41 mm. 287 parts. Zero routed tracks, vias or planes. 1,091 PCB connections await copper.

M1 FPGA board in 3D



Nominal AX3 and NVT4858 bodies show placement envelopes. Physical mating and clearance remain unqualified.

Simpler ASIC connections

M1 removes all 15 ASIC bus buffers and 56 parts overall. The component count falls from 343 to 287.

The original same-ASIC LDO and ECP5 reference circuits connect their digital signals directly.

M1 retains the outgoing 33 Ω damping arrays, passive defaults, boot holds and stimulation permission logic.

Removing buffers also removes their powered-off isolation. Controlled power sequencing and interface qualification remain necessary.

Gerald's board-design guidance

Place repeated circuits clearly, following signal flow and short local power loops.

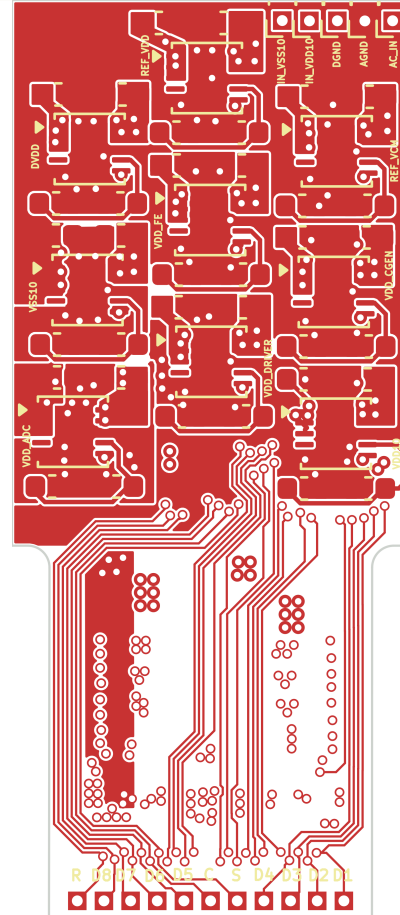
Match total clock/data timing within each related I/O group. Shorten detours before tuning.

Keep continuous ground references and deliberate return connections at layer changes. Place bypass capacitors at their ICs.

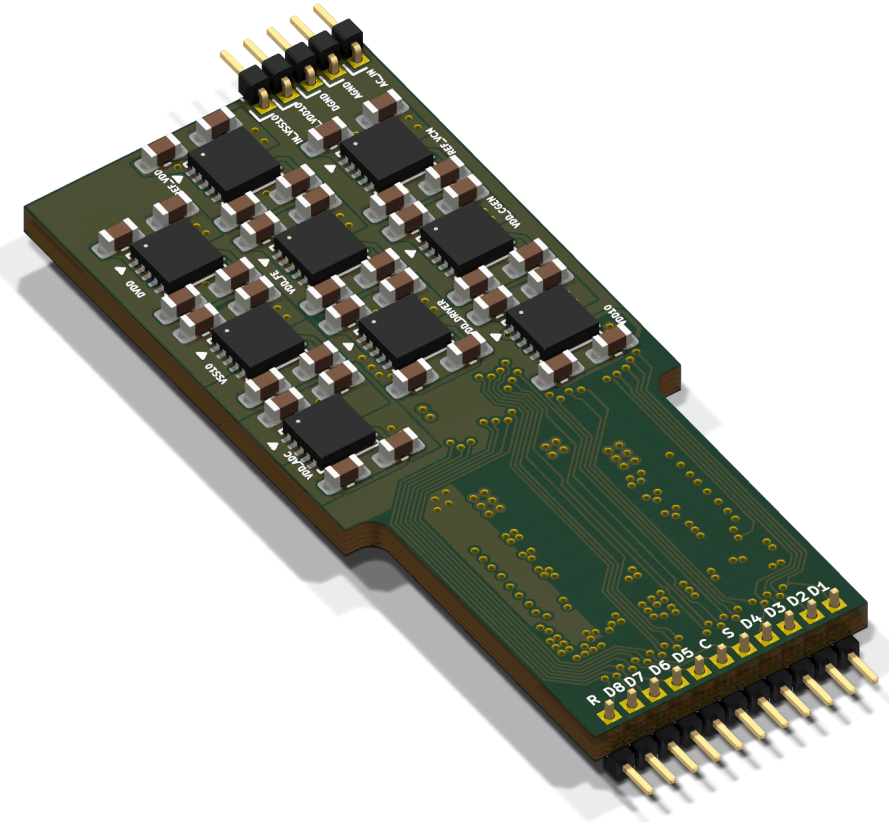
Use manufacturable FPGA escapes. Gerald's 1–2, 1–3 and 1–4 blind-via spans need a fabricator-approved construction.

Keep necessary functions in a compact placement, with bottom mezzanine connectors and a clear overall schematic.

Existing ASIC LDO board



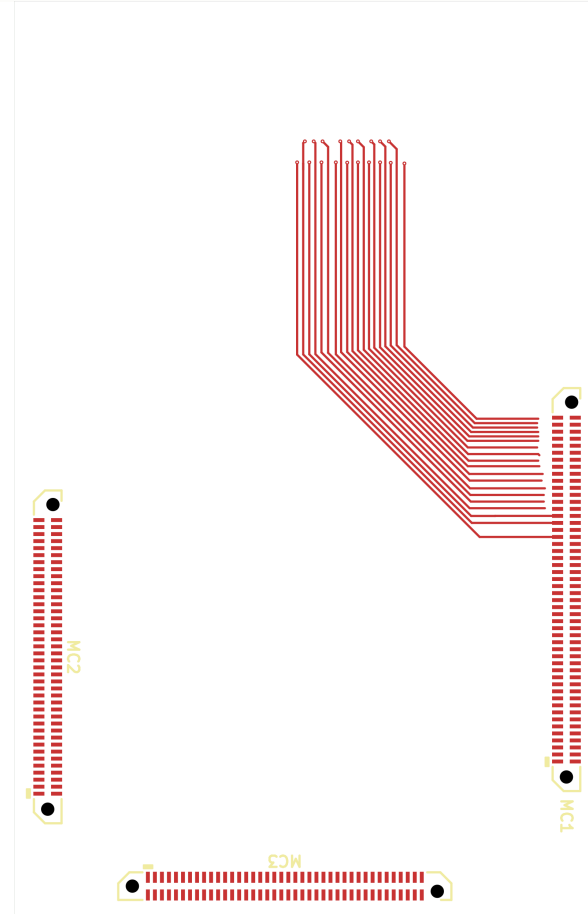
Actual native board layout



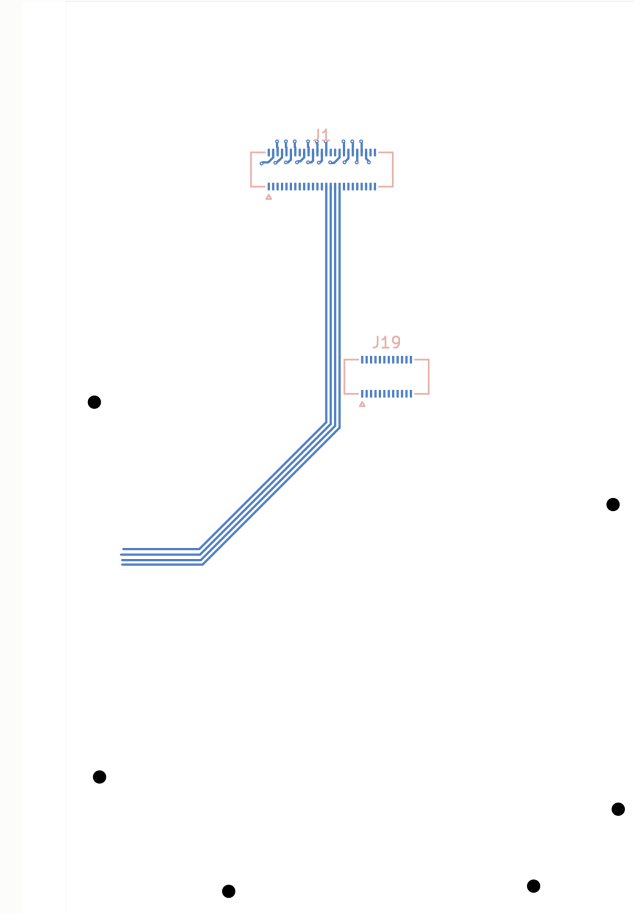
Native KiCad 3D view

Nine regulator/reference stages. J1/J19 housing models are absent. J19 has no assigned nets. Installed population remains unverified.

Received direct-carrier layout



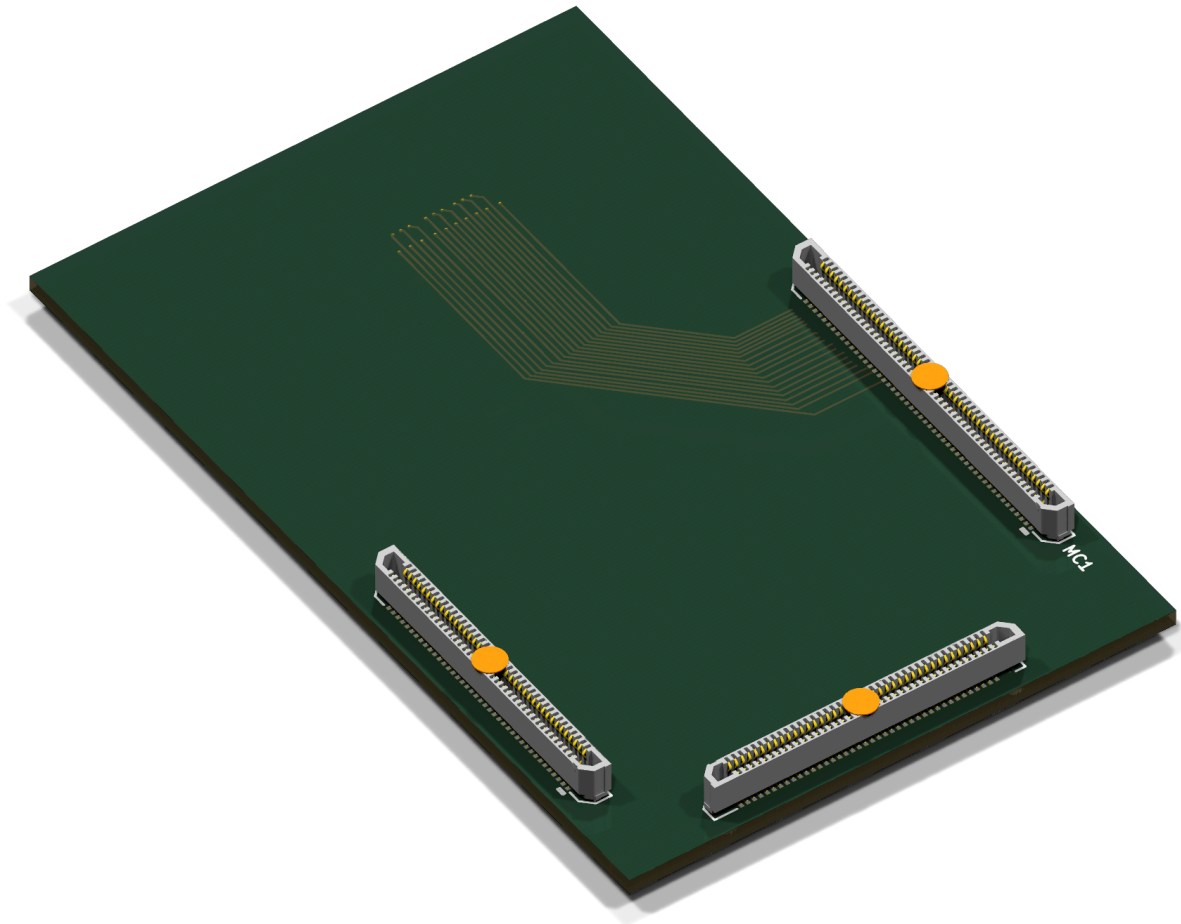
New Desktop carrier, top layout



New Desktop carrier, underside layout

Board_Howard draft: 38 DRC errors, 21 dangling-track warnings and 20 assigned open connections. Compare with saved Step 8.

Received carrier in 3D



Native preview includes the Samtec XEM sockets. Hirose J1/J19 housings are missing. Seating and assembled fit remain open.

Carrier routing status

Native check	Applied Step 8 baseline	New Desktop draft
Physical DRC findings	0	38 errors + 21 warnings
Assigned open connections	0	20
Parity warnings	204	204
Timing acceptance	Open	Open
Power / startup circuit	Pending	Pending

Step 8 returned data/clock geometry: 48.547–63.830 mm, 15.283 mm spread. This measurement belongs to the saved baseline.

Next review decisions

Confirm the FPGA clock path, oscillator tolerance and SPI boot behavior for the selected parts.

Review the compact placement and carrier routing with Gerald, then route against an explicit timing and return-path plan.

Close ASIC pad limits, per-rail loads, power order and shared-control loading before operating the direct interface.

Check simultaneous connector mating, actual component bodies and the manufacturer's stackup before fabrication.

Build the matched FPGA/host workflow and test one ASIC before assessing eight-chip operation.